Simplifying Graph Convolutional Networks

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OBJECTIVES OVERVIEW

- GCNs can be simplified by removing nonlinearities and collapsing weight matrices into feature propagation followed by a linear classifier
- SGC works as a fixed low-pass filter that smooths node features over neighbors
- Model becomes faster and still comparable in accuracy to GNN models.
- Datasets: Cora, Citeseer, Pubmed, and Reddit

METHODS

GCN

$$H^{(k)} = \mathrm{ReLU}(SH^{(k-1)}\theta^{(k)})$$

↓ remove nonlinearity + collapse weights

Feature Propagation

$$\bar{X} = S^K X$$

↓ simple linear classifier

SGC

$$\hat{Y} = \mathrm{softmax}(\bar{X}\theta)$$

EVALUATION

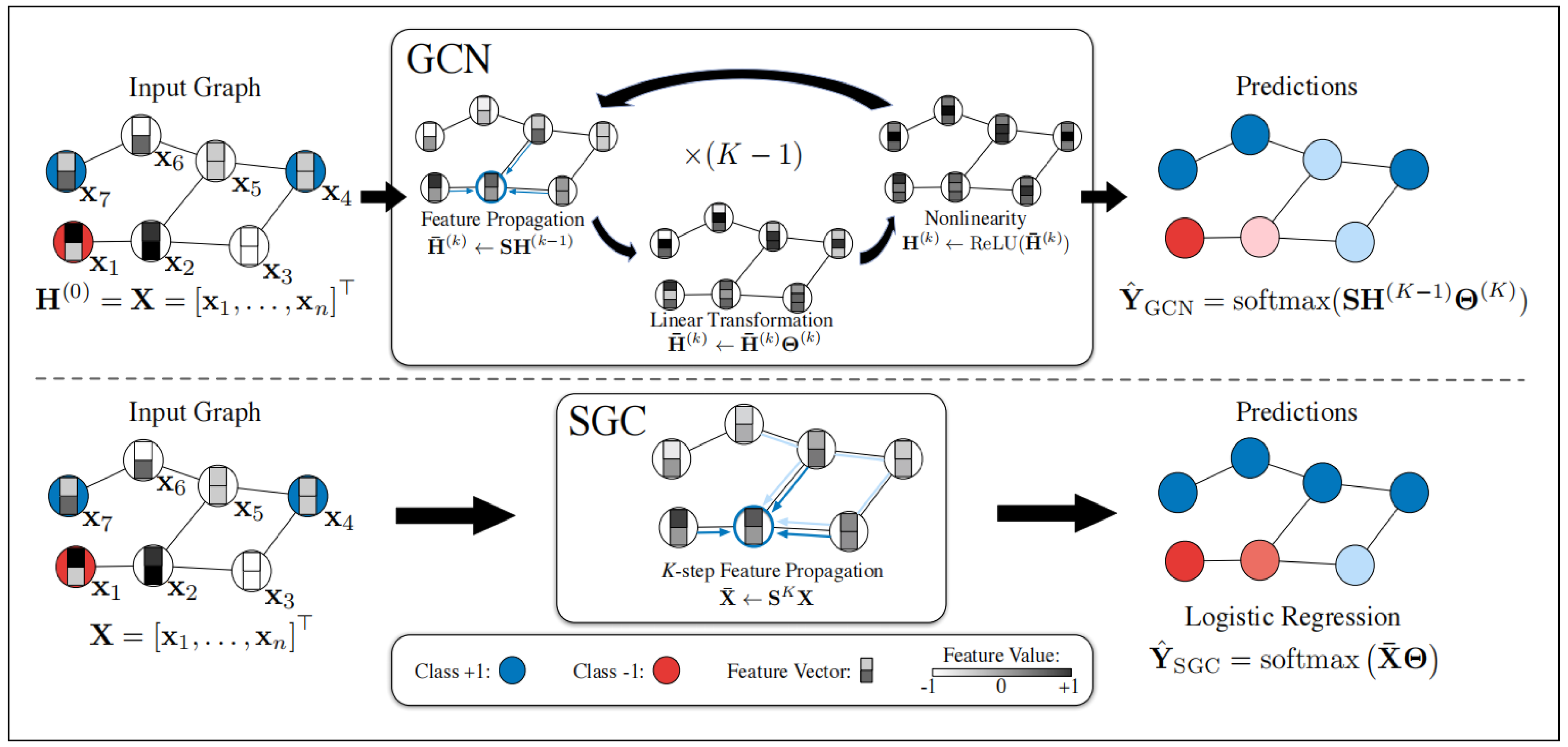
$$Accuracy = \frac{1}{N}\sum 1(\hat{y}_i = y_i)$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

MODEL ARCHITECTURE



RESULTS

Performance and Training Time

Model	Cora		Citeseer		Pubmed		Reddit	
	Acc(%)	Time	Acc(%)	Time	Acc(%)	Time	Micro-F1(%)	Time
SGC(ours)	82.0 ± 0.0	0.2s	71.8 ± 0.0	0.2s	78.9 ± 0.0	0.2s	94.6	5.7s
GCN	81.4 ± 0.9	0.8s	68.9 ± 1.2	0.9s	79.1 ± 0.5	1.2s	OOM	OOM
FastGCN	82.1 ± 1.2	1.4s	68.3 ± 1.0	2.3s	76.0 ± 0.8	2.8s	90.4	1581.9s

SGC achieves the best trade-off between efficiency and performance across all datasets.

Citation Networks and Reddit Training Setup

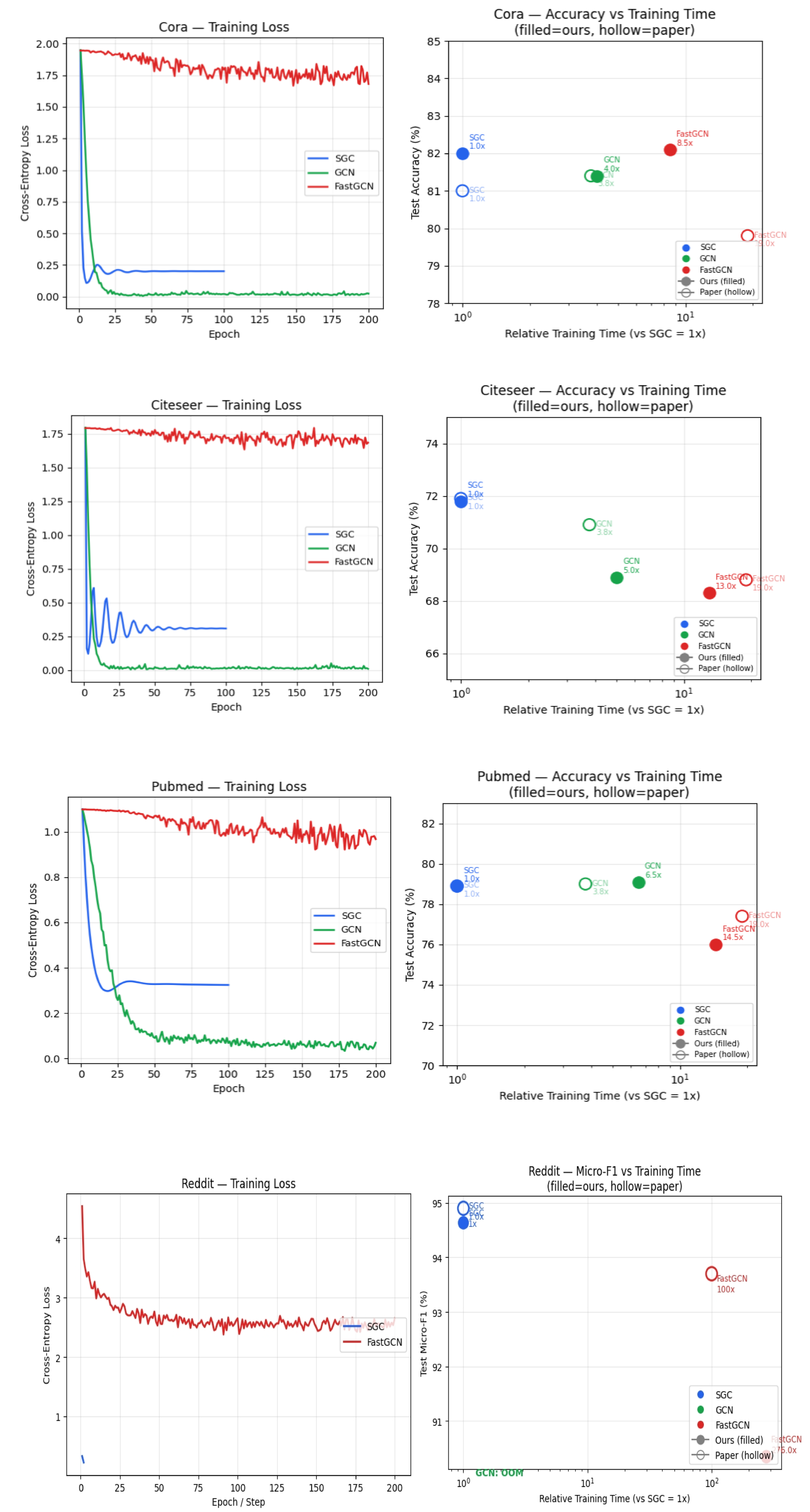
Model	Architecture	Epochs	Optimizer	Learning Rate	Weight Decay	Notes
SGC(ours, C)	Linear(K=2)	100	Adam	0.2	$10^{-10} - 10^{-2}$	Precompute $S^K X$
SGC(ours, R)	Linear(K=2)	20 steps	L-BFGS	1.0	0	Precompute $S^K X$; ; train on subgraph
GCN(C)	2-layer, 64 hidden	200	Adam	0.01	5×10^{-4}	Dropout 0.5
GCN(R)	2-layer, 64 hidden	-	Adam	0.01	5×10^{-4}	Full-batch training (OOM on Reddit)
FastGCN(C)	2-layer, 64 hidden	200	Adam	0.01	5×10^{-4}	Sampling(400 nodes for Cora/Citeseer, 4000 nodes for Pubmed)
FastGCN(R)	2-layer, 64 hidden	200	Adam	0.01	5×10^{-4}	Importance sampling (batch training)

FUTURE WORK

- Extend SGC to diverse downstream tasks (e.g., text classification, relation extraction) to evaluate its generalization and scalability in real-world scenarios.
- Extend SGC with adaptive propagation to relax the fixed-K assumption and capture varying receptive fields across nodes (if time permits).

CONCLUSIONS

Best for simple, large-scale Node Classification on Homophilous Graphs



REFERENCES

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